

Name:

Points:

4 / 20

6

Databasutveckling och lagring — Omtenta Normalisering

2025-05-26

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. You will be presented with a database schema that isn't normalized. There are **6 normalization problems** in the given schema.
6. For each problem you need to write shortly (2 short sentences) about:
 - what the problem is
 - in which way it breaks normalization
 - how to fix it

Example:

The column X in the table Y breaks normalization by If we ..., we can solve the issue.

7. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Grading

Problems correctly recognized and solved	Grade
0 - 2	IG
3 - 5	G
6	VG

1 Intro

The local university is migrating to a new system, and they assigned their intern to design the new database schema. Now they've noticed there seems to be some problems with normalization, and they asked us to help them solve them.

As it turns out, the problems aren't due to the intern's lack of knowledge, the intern was in fact much more knowledgeable than it first seemed. A goodbye note was found at their empty desk, where the intern expresses their deep hate for the University, as well as admits to leaving **6 normalization errors** in the design as a *middle-finger* to the institution they so much despise.

It's up to us - as proud alumni of the University - to come to their rescue by normalizing the database.

2 Tables & sample rows

2.1 Students

StudentID	FirstName	LastName	Phones
1	Alice	Andersson	0701234567,0727654321
2	Bob	Berg	0731112222

2.2 Enrollments

StudentID	CourseID	CourseName	Credits	EnrollmentDate	Grade
1	CS101	Intro to Programming	5	2025-01-15	A
1	DB201	Databases	6	2025-01-15	B
2	CS101	Intro to Programming	5	2025-01-15	C

2.3 InstructorCourses

InstructorID	CourseID	InstructorName	InstructorPhone
10	CS101	Dr Smith	0709876543
11	DB201	Dr Karlsson	0708765432

2.4 Courses

CourseID	CourseName	DeptID	DeptName	DeptHead
CS101	Intro to Programming	D1	Computer Science	Prof Adams
DB201	Databases	D1	Computer Science	Prof Adams
MAT100	Calculus I	D2	Mathematics	Prof Brown

2.5 Rooms

RoomID	RoomNumber	BuildingID	BuildingName	BuildingAddress	Capacity
R101	101	B1	Main Hall	Campus Road 1	40
R202	202	B2	Science Center	Science Way 5	35

2.6 Books

ISBN	Title	PublisherName	PublisherAddress
978-1-11111-111-1	Database Systems	TechBooks	123 Tech Ave
978-1-22222-222-2	Intro to Programming	EduPress	45 Learning St

```
CREATE TABLE Students (
  StudentID INT PRIMARY KEY,
  FirstName VARCHAR(50),
  LastName VARCHAR(50),
  Phones VARCHAR(255)
);
```

```
CREATE TABLE Courses (
  CourseID VARCHAR(10) PRIMARY KEY,
  CourseName VARCHAR(100),
  DeptID VARCHAR(10),
  DeptName VARCHAR(100),
  DeptHead VARCHAR(100)
);
```

```
CREATE TABLE Enrollments (
  StudentID INT,
  CourseID VARCHAR(10),
  CourseName VARCHAR(100),
  Credits TINYINT,
  EnrollmentDate DATE,
  Grade CHAR(1),
  PRIMARY KEY (StudentID, CourseID)
);
```

```
CREATE TABLE Rooms (
  RoomID VARCHAR(10) PRIMARY KEY,
  RoomNumber VARCHAR(10),
  BuildingID VARCHAR(10),
  BuildingName VARCHAR(100),
  BuildingAddress VARCHAR(150),
  Capacity INT
);
```

```
CREATE TABLE InstructorCourses (
  InstructorID INT,
  CourseID VARCHAR(10),
  InstructorName VARCHAR(100),
  InstructorPhone VARCHAR(15),
  PRIMARY KEY (InstructorID, CourseID)
);
```

```
CREATE TABLE Books (
  ISBN VARCHAR(17) PRIMARY KEY,
  Title VARCHAR(150),
  PublisherName VARCHAR(100),
  PublisherAddress VARCHAR(150)
);
```

-- followed by the INSERT statements
-- that match the rows above

2.1 bryter 1NF alla rader måste vara unika

1

2 Alice Adersson 0727654321

3 Alice Adersson 0701234567

2.2 den här tabellen ^{Grader} CourseName bryter bara på CourseID
den bryter mot 2NF

2.3 den bryter 3NF och 2NF för att InstructorPhone bara
~~bryter~~ bryter på InstructorName och ingen av de bryter på
CourseID man måste göra 2 tabeller en för InstructorID
och en för CourseID

2.5 bryter 3NF för att Alla nonc primary key ska bryta på
primary key och inte på varandra ³ man måste göra 2 tabeller
en för RoomID och en för BuildingID

4 Normalization problems

4.1 1st normalization problem

students

X Alla rader måste vara unika så man måste göra en till rad för Alice Andersson med 1 telefon
den telefon

4.2 2nd normalization problem

X 2.1 bryter 1NF Alice andersson måste ha 2 rader

4.3 3rd normalization problem

✓ 2.2 bryter 2NF man måste göra 2 tabeller av data, en student ID och en course ID

4.4 4th normalization problem

✓ 2.3 bryter 2NF och 3NF det måste finnas 2 tabeller, en för instructor ID och en course ID som kopplas ihop med foreign key

4.5 5th normalization problem

✓ 2.5 bryter 3NF måste göras 2 tabeller Room ID och Building ID så de kopplas ihop

4.6 6th normalization problem

✓ 2.4 bryter 2NF måste göras 2 tabeller course ID och Room ID